

Saber Ayatollahi, CFA, MSc

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SUMMARY

Senior Associate, Investment Engineering & Analytics candidate with ~7 years building quantitative investment and risk analytics for institutional portfolios. At Moody's Analytics, designs and delivers production Python analytics pipelines and multi-asset models supporting portfolio managers; at Ortec Finance, ran systematic asset-only and surplus portfolio optimization on VaR/CVaR with stochastic macro scenario generators, risk decomposition, and factor-style attribution. CFA charterholder; dual MSc (Financial Modelling, Engineering); fluent in Python and SQL.

CORE SKILLS

Python · pandas · NumPy · SciPy • SQL · PostgreSQL • Systematic macro strategy analytics • Factor-style risk attribution & decomposition • Portfolio optimization (VaR / CVaR) • Stochastic & Monte Carlo modelling • Production analytics pipelines · Git · CI/CD • Agentic AI development (Claude Code, Cursor)

EXPERIENCE

Moody's Analytics

Asst. Director, Modelling Services | Toronto, May 2023 – Present

Production Models & Portfolio Analytics

- Hold delegated sign-off authority on valuation, sensitivity, and portfolio analytics outputs for multi-asset institutional portfolios ranging \$5-25bn per engagement, cumulatively covering ~\$50bn in assets.
- Validate derivatives pricing outputs across rates, FX, and inflation, and review aggregation logic translating security-level exposures into portfolio-level risk and return metrics consumed by portfolio managers.
- Review curve construction, spread calibration, and cross-asset interactions prior to production release within a formal model governance framework – escalating outputs that are mathematically defensible but economically unsupported.

Data Pipelines & Engineering

- Led design and implementation of an enterprise multi-asset cash flow projection engine supporting base, stress, and reverse-stress macro scenarios across multi-year horizons.
- Re-engineered manual spreadsheet workflows into scalable, auditable Python analytics pipelines with logging, validation, and reproducibility controls.
- Architected configurable time-bucketed analytics (T+1 through multi-year) supporting funding, allocation, and refinancing decisions for live institutional portfolios.

AI-Augmented Development & Research Partnership

- Built agentic AI development workflows (Claude Code, Cursor IDE) automating code generation, validation scaffolding, anomaly detection, and documentation – reducing development cycle time an estimated 30-40% on comparable modules.
- Partner with product owners and client investment teams to translate analytical requirements into production-grade Python; mentor junior colleagues on coding standards and review practices.

Client Service Specialist | Toronto, May 2022 – May 2023

- Onboarded U.S. and Canadian pension funds, asset managers, and consulting firms onto the PFaroe multi-asset analytics platform; validated model outputs post-deployment.
- Scoped client analytical requirements into structured Product Owner requests, bridging investment teams and development to ship analytics enhancements into production.

Ernst & Young (EY)

Senior Consultant, Financial Services Risk | Toronto, Sep 2021 – Apr 2022

- Delivered IFRS 17 and IFRS 9 technology-transformation projects for Canadian and international insurers, aligning data, finance, actuarial, and IT workstreams.
- Contributed to governance documentation and analytics design supporting regulatory-grade financial reporting.

Ortec Finance Canada

Consultant, ALM & Risk | *Toronto, Feb 2019 – Sep 2021*

- Performed systematic asset-only and asset-liability (surplus) portfolio optimization on VaR and CVaR using Ortec's GLASS platform, with risk decomposition, contribution-to-risk, and near-efficient-frontier robustness analysis.
- Built and interpreted stochastic macro scenario generators driving multi-asset return, inflation, rate, and currency dynamics for institutional portfolios.
- Conducted factor- and asset-class-level risk attribution and hedging analysis (rates, inflation, FX) for international institutional clients.
- Advised pension fund and institutional clients on Strategic and Tactical Asset Allocation via stress testing, sensitivity analysis, and scenario-based research.
- Led ALM study for the University Pension Plan merger (three single-employer plans into a new jointly-sponsored plan), developing and validating models capturing funding-policy dynamics, duration, currency, inflation, and leverage overlays.
- Presented research findings and portfolio recommendations to client investment committees on-site.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) – CFA Institute, 2024
- MSc, Financial Modelling – Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) – Western University, 2016